What is the name of element with atomic number 105

(a) Kurchatovium

(b) Dubnium

(c) Nobelium

(d) Holmium

Answer: (b) Dubnium

Explanation: Atomic number 105 is dubnium (Db).

222. Electrons in a paramagnetic compound are

(a) Shared

(b) Unpaired

(c) Donated

(d) Paired

Answer: (b) Unpaired

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

223. Which of the following pairs involves isoelectronic ions

(a) Mn3+and Fe2+

(b) Mn2+ and Fe3+

(c) Cr3+ and Fe2+

(d) Fe2+ and Co2+

Answer: (b) Mn2+ and Fe3+

Explanation: The correct choice is Mn2+ and Fe3+. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

224. Which of the following is paramagnetic

(a) Ni++

(b) Cu+

(c) Zn++

(d) Sc+++

Answer: (a) Ni++

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

225. The electronic configuration of chromium is

(a) [Ne]3s23p63d4,4s2

(b) [Ne]3s23p63d5,4s1

(c) [Ne]3s23p63d6,4s1

(d) [Ne]3s23p53d5,4s2

Answer: (b) [Ne]3s23p63d5,4s1

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

226. Electronic configuration of Cu(Z=29) is

(a) [Ar]3d94s2

(b) [Ar]3d104s1

(c) [Ar]3d54s2

(d) [Ar]3d64s2

Answer: (b) [Ar]3d104s1

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

227. Ce-58 is a member of

(a) s-block

(b) p-block

(c) d-block

(d) f-block

Answer: (d) f-block

Explanation: The correct choice is f-block. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

228. How many unpaired electrons are there in

(a) 2

(b) 4

(c) 5

(d) 0

Answer: (a) 2

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

229. The main reason for larger number of oxidation states exhibited by the actinoids than the corresponding lanthanoids is

(a) Lesser energy difference between 5f and 6d orbitals than between 4f and 5d orbitals

(b) Larger atomic size of actinoids than the lanthanoids

(c) More energy difference between 5f and 6d orbitals than between 4f and 5d orbitals

(d) Greater reactive nature of the actinoids than the lanthanoids

Answer: (a) Lesser energy difference between 5f and 6d orbitals than between 4f and 5d orbitals

Explanation: Assign oxidation numbers from charge balance and the accessible ns/(n-1)d valence electrons of the transition element.

230. Four successive members of the first row transition elements are listed below with their atomic numbers. Which one of them is expected to have the highest third ionization enthalpy

(a) Vanadium (Z = 23)

(b) Chromium (Z = 24)

(c) Iron (Z = 26)

(d) Manganese (Z = 25)

Answer: (d) Manganese (Z = 25)

Explanation: Compare the stability of the resulting electronic configurations as well as effective nuclear charge; half-filled and fully filled subshells often produce anomalies.

231. The aqueous solution containing which one of the following ions will be colourless

(a) Sc^{3+}
(b) Fe^{2+}
(c) Ti^{3+}
(d) Mn^{2+} (Atomic number Sc = 21, Fe = 26, Ti = 22, Mn = 25)

Answer: (a) Sc^{3+}

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

232. Which of the following trivalent ion has the largest atomic radii in the lanthanide series

(a) La
(b) Ce
(c) Pm
(d) Lu

Answer: (a) La

Explanation: Across the lanthanoids, 4f electrons shield the nuclear charge poorly, so effective nuclear charge rises and atomic/ionic radii generally decrease.

233. Which of the following does not have valence electron in 3d-subshell

(a) Fe (III)
(b) Mn (II)
(c) Cr (I)
(d) P (0)

Answer: (d) P (0)

Explanation: The correct choice is P (0). This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

234. Among the following pairs of ions, the lower oxidation state in aqueous solution is more stable than the other in

(a) Tl^{+}, Tl^{3+}
(b) Cu^{+}, Cu^{2+}
(c) Cr^{2+}, Cr^{3+}
(d) V^{2+}, VO^{2+}

Answer: (a) Tl^{+}, Tl^{3+}

Explanation: Assign oxidation numbers from charge balance and the accessible ns/(n-1)d valence electrons of the transition element.

235. The lanthanide contraction is responsible for the fact that

(a) Zr and Y have about the same radius
(b) Zr and Nb have similar oxidation state

(c) Zr and Hf have about the same radius

(d) Zr and Zn have the same oxidation sate

Answer: (c) Zr and Hf have about the same radius

Explanation: Across the lanthanoids, 4f electrons shield the nuclear charge poorly, so effective nuclear charge rises and atomic/ionic radii generally decrease.

236. Which of the following factors may be regarded as the main cause of lanthanide contraction

(a) Poor shielding of one of 4f electron by another in the subshell

(b) Effective shielding of one of 4f electrons by another in the subshell

(c) Poorer shielding of 5d electrons by 4f electrons

(d) Greater shielding of 5d electron by 4f electrons

Answer: (a) Poor shielding of one of 4f electron by another in the subshell

Explanation: Across the lanthanoids, 4f electrons shield the nuclear charge poorly, so effective nuclear charge rises and atomic/ionic radii generally decrease.

237. Which of the following have maximum number of unpaired electrons

(a) Fe^{3+}

(b) Fe^{2+}

(c) Co^{2+}

(d) Co^{3+}

Answer: (a) Fe^{3+}

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d^0 and d^{10} species are diamagnetic.

238. Transition metals show paramagnetism

(a) Due to characteristic configuration

(b) High lattice energy

(c) Due to variable oxidation states

(d) Due to unpaired electrons

Answer: (d) Due to unpaired electrons

Explanation: The correct choice is Due to unpaired electrons. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

239. Which of the following pairs of elements cannot form an alloy

(a) Zn,Cu

(b) Fe,Hg

(c) Fe,C

(d) Hg,Na

Answer: (b) Fe,Hg

Explanation: The correct choice is Fe,Hg. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

240. Which belongs to the actinides series

(a) Ce
(b) Cf
(c) Ca
(d) Cs

Answer: (b) Cf

Explanation: Actinoids involve 5f orbitals; the small energy differences among 5f, 6d and 7s levels allow several oxidation states and extensive complex formation.

241. Effective magnetic droment of Sc+3ion is

(a) 1.73
(b) 0
(c) 5.92
(d) 2.83
(e) 3.87

Answer: (b) 0

Explanation: Sc3+ has the configuration [Ar]3d0, so it has no unpaired electrons. Therefore μ = √[n(n+2)] = 0 BM.